

```
In [1]: import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
In [3]: # Load dataset
df = pd.read_csv("data/user_data.csv")
print("csv file read")
print(df.head())
print(df.shape)
```

```
csv file read
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
(19, 5)
```

```
In [4]: # Remove unnecessary column
df = df.drop(columns=['User ID'])

# Encode categorical variable: Gender (Male=1, Female=0)
le = LabelEncoder()
df['Gender'] = le.fit_transform(df['Gender'])

print(df.head())
```

	Gender	Age	EstimatedSalary	Purchased
0	1	19	19000	0
1	1	35	20000	0
2	0	26	43000	0
3	0	27	57000	0
4	1	19	76000	0

```
In [5]: # Separate features (X) and target (y)
X = df[['Gender', 'Age', 'EstimatedSalary']].values
y = df['Purchased'].values
```

```
In [6]: # Apply feature scaling
scaler = StandardScaler()
X = scaler.fit_transform(X)
```

```
In [7]: # Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
In [8]: # Create and train Logistic Regression model
model = LogisticRegression()
model.fit(X_train, y_train)
```

Out[8]:

```
▼ LogisticRegression ⓘ ?  
LogisticRegression()
```

In [9]:

```
# Predict on test set  
predictions = model.predict(X_test)
```

In [10]:

```
# Evaluate performance  
accuracy = accuracy_score(y_test, predictions)  
print("Accuracy:", accuracy)  
print("\nConfusion Matrix:")  
print(confusion_matrix(y_test, predictions))  
print("\nClassification Report:")  
print(classification_report(y_test, predictions))
```

Accuracy: 1.0

Confusion Matrix:
[[4]]

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	4
accuracy			1.00	4
macro avg	1.00	1.00	1.00	4
weighted avg	1.00	1.00	1.00	4

C:\Users\222-G\anaconda3\Lib\site-packages\sklearn\metrics_classification.py:407: UserWarning: A single label was found in 'y_true' and 'y_pred'. For the confusion matrix to have the correct shape, use the 'labels' parameter to pass all known labels.
warnings.warn(

In [11]:

```
# Display sample predictions: Age, Salary, Actual, Predicted  
# Inverse-transform scaled X_test to recover original Age and Salary values  
X_test_original = scaler.inverse_transform(X_test)  
  
results = pd.DataFrame({  
    'Age': X_test_original[:, 1].astype(int),  
    'Salary': X_test_original[:, 2].astype(int),  
    'Actual': y_test,  
    'Predicted': predictions  
})  
  
print(results.head())
```

	Age	Salary	Actual	Predicted
0	19	19000	0	0
1	27	58000	0	0
2	26	52000	0	0
3	35	20000	0	0